



# 9886 - Lyman-alpha tomography of a proto-cluster with a luminous quasar in the early universe

Cycle: 5, Proposal Category: GO

## INVESTIGATORS

| <i>Name</i>                             | <i>Institution</i>                                |
|-----------------------------------------|---------------------------------------------------|
| <b>Prof. Anna-Christina Eilers (PI)</b> | <b>Massachusetts Institute of Technology</b>      |
| Prof. Robert Andrew Simcoe (CoI)        | Massachusetts Institute of Technology             |
| Dr. Jorrit Matthee (CoI) (ESA Member)   | Institute of Science and Technology Austria       |
| Dr. Rohan Naidu (CoI)                   | Massachusetts Institute of Technology             |
| Dr. Frederick Davies (CoI) (ESA Member) | Max Planck Institute for Astronomy                |
| Prof. Joseph Hennawi (CoI)              | University of California - Santa Barbara          |
| Dr. Minghao Yue (CoI)                   | University of Arizona                             |
| Dr. Rongmon Bordoloi (CoI)              | North Carolina State University                   |
| Dr. Ruari Mackenzie (CoI) (ESA Member)  | Ecole Polytechnique Federale de Lausanne          |
| Dr. Daichi Kashino (CoI)                | National Astronomical Observatory of Japan (NAOJ) |

## OBSERVATIONS

| <i>Folder</i>      | <i>Observation</i> | <i>Label</i>                                  | <i>Observing Template</i>        | <i>Science Target</i> |
|--------------------|--------------------|-----------------------------------------------|----------------------------------|-----------------------|
| Observation Folder |                    |                                               |                                  |                       |
|                    | 2                  | J0148_PA200_right_v2<br>, J0148_PA200_left_v2 | NIRSpec MultiObject Spectroscopy | (1) catalog_J0148     |

## ABSTRACT

The ultraviolet radiation of luminous high-redshift quasars within the Epoch of Reionization carves out large bubbles of highly ionized intergalactic gas in the quasars' environments, making the otherwise opaque Lyman-alpha (Ly $\alpha$ ) forest highly transparent at redshifts near to the systemic redshift of the quasar, which is known as the proximity effect. In this proposal we aim to leverage the unique capabilities of JWST's NIRSpec/MSA to

## JWST Proposal 9886 (Created: Friday, March 13, 2026, 1:01:17PM Eastern Standard Time) - Overview

observe spectroscopically confirmed, bright galaxies in the background of a luminous quasar at  $z \sim 6$  at close projected distances, in order to tomographically map the quasar's "light echo", i.e. the regions ionized by the quasar by means of Ly $\alpha$  tomography. These observations will uniquely reveal the geometry of the quasars' ionization cone, and constrain their opening angles and obscuration fractions. Additionally, these ionized regions encode the quasar's radiative history, and thus provide a model-independent constraint on the quasar's age based on the light travel time, which will enable new insights on the puzzle of the early growth of supermassive black holes.

At the same time, the quasar resides in one of the most impressive proto-clusters known to date with more than 47 [OIII]-emitting galaxies in its vicinity. Thus, the highly ionized quasar environment provides a unique laboratory to study the intrinsic spectra of high-redshift galaxies without absorption by the otherwise neutral IGM. By studying the unattenuated line profiles of the Ly $\alpha$  emission lines of galaxies, our proposed observations will also measure their escape fractions and reveal the protagonists that reionize our universe.

### **OBSERVING DESCRIPTION**

We target bright ( $m_{AB} < 27$  mag in F115W) galaxies in a field around a luminous quasar, i.e. J0148+0600 at  $z=5.99$  using NIRSpec/MSA. The targeted galaxies are spectroscopically confirmed by previous NIRCам WFSS observations at redshifts  $5.3 < z < 7$ .

In order to obtain a spatially well distributed sample of galaxies required for our science goals we need two NIRSpec/MSA configurations per field. We choose the medium resolution grating + filter G140M/F070LP, which provides continuous wavelengths coverage between 0.70–1.27  $\mu$ m with a resolution of  $R \sim 1000$ . For Lyman-alpha at the quasars' redshifts at approximately 0.85  $\mu$ m, the resolution is slightly lower, i.e.  $R \sim 700$ , corresponding to a pixel scale of  $v \sim 420$  km/s. This resolution will enable us to resolve flux transmission within the Lyman-alpha forest of background galaxies with approximately 2-10 spectral pixels (depending on the pathlengths with high flux transmission), as well as de-blend the low EW of the UV emission lines and Ly $\alpha$  emission of high-redshift galaxies.

We require 7.3 hours on source to obtain  $SNR \sim 2$  per resolution element for a galaxy of  $m_{AB} \sim 26.5$  mag, which corresponds to the 25% brightest galaxies in our sample. This will enable us to obtain a  $>3$  sigma detection of the transverse proximity effect in four independent radial and azimuthal bins, allowing constraints on the extent and geometry of the quasars' ionized regions.

In order to contain the data rate, we chose the NRSIRS2 readout pattern and 20 groups per integration, such that the total exposure time per integration is below the recommended value of 1500s. We use the recommended dither pattern with 3 nods in the shutter slitlets. These observations are duplicated six times, in order to achieve the required exposure time per MSA configuration.

We will obtain target confirmation images, with 5 groups per integration, which will result in  $>3$  sigma detections even for the faintest sources on slits ( $m_{AB} \sim 28$  mag).

# Proposal 9886 - Targets - Lyman-alpha tomography of a proto-cluster with a luminous quasar in the early universe

| Fixed Targets | #                           | Name          | Target Coordinates                                                                | Targ. Coord. Corrections | Miscellaneous |
|---------------|-----------------------------|---------------|-----------------------------------------------------------------------------------|--------------------------|---------------|
|               | (1)                         | catalog_J0148 | RA: 01 48 38.0654 (27.1586058d)<br>Dec: +06 00 23.64 (6.00657d)<br>Equinox: J2000 |                          |               |
|               | Comments:<br>Description=[] |               |                                                                                   |                          |               |

# Proposal 9886 - Observation 2 - Lyman-alpha tomography of a proto-cluster with a luminous quasar in the early universe

|                   |                                                                                                                                                                                                                                                                                                                                                                 |                        |                                                                                                                      |                   |                                                               |                          |                              |                                    |                            |                     |                              |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------|--------------------------|------------------------------|------------------------------------|----------------------------|---------------------|------------------------------|
| Observation       | Proposal 9886, Observation 2: J0148_PA200_right_v2, J0148_PA200_left_v2                                                                                                                                                                                                                                                                                         |                        |                                                                                                                      |                   |                                                               |                          |                              |                                    |                            |                     | Fri Mar 13 18:01:17 GMT 2026 |
|                   | Diagnostic Status: Warning<br>Observing Template: NIRSpec MultiObject Spectroscopy                                                                                                                                                                                                                                                                              |                        |                                                                                                                      |                   |                                                               |                          |                              |                                    |                            |                     |                              |
| Diagnosics        | (J0148_PA200_right_v2, J0148_PA200_left_v2 (Obs 2)) Warning (Form): Config c1 : J0148_PA200_right_v2 (#1) has 2 primary slits affected by failed closed shutters.<br>(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.<br>(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                        |                                                                                                                      |                   |                                                               |                          |                              |                                    |                            |                     |                              |
| Fixed Targets     | #                                                                                                                                                                                                                                                                                                                                                               | Name                   | Target Coordinates                                                                                                   |                   |                                                               | Targ. Coord. Corrections |                              |                                    | Miscellaneous              |                     |                              |
|                   | (1)                                                                                                                                                                                                                                                                                                                                                             | catalog_J0148          | RA: 01 48 38.0654 (27.1586058d)<br>Dec: +06 00 23.64 (6.00657d)<br>Equinox: J2000<br><br>Comments:<br>Description=[] |                   |                                                               |                          |                              |                                    |                            |                     |                              |
| Acquisition       | #                                                                                                                                                                                                                                                                                                                                                               | Reference Star Bin     | Target                                                                                                               | Filter            | MSA Configuration                                             | Readout Pattern          | Groups/Int                   | Integrations/Exp                   | Total Integrations         | Total Exposure Time | Optional ETC ID              |
|                   | 1                                                                                                                                                                                                                                                                                                                                                               |                        | SAME                                                                                                                 | F140X             | Auto Acq MSA Config                                           | NRSRAPID                 | 3                            | 1                                  | 4                          | 171.788             |                              |
|                   | 2                                                                                                                                                                                                                                                                                                                                                               |                        | SAME                                                                                                                 | F140X             | Auto Acq MSA Config                                           | NRSRAPID                 | 3                            | 1                                  | 4                          | 171.788             |                              |
| Template          | TA Method                                                                                                                                                                                                                                                                                                                                                       | HFF Readout Mode       | Obtain Confirmation Images                                                                                           | Science Aperture  | Primary Candidate List                                        | Filler Candidate List    | Spectral Overlap Map         |                                    | Spectral Overlap Threshold |                     |                              |
|                   | MSATA                                                                                                                                                                                                                                                                                                                                                           | false                  | No                                                                                                                   | MSA Center        | primary (118 sources)                                         | secondary (35 sources)   | jwst-nirspec-g140m           |                                    | 1.5                        |                     |                              |
| Reference Stars   |                                                                                                                                                                                                                                                                                                                                                                 |                        |                                                                                                                      |                   |                                                               |                          |                              |                                    |                            |                     |                              |
| Spectral Elements | #                                                                                                                                                                                                                                                                                                                                                               | Exposure Specification | MSA Configuration                                                                                                    | Nod Pattern       | Pointing                                                      | Aperture PA              | Dispersion Offset (Shutters) | Cross-Dispersion Offset (Shutters) | Total Dithers              | Total Integrations  | Total Exposure Time          |
|                   | 1                                                                                                                                                                                                                                                                                                                                                               | 1<br>(G140M/F070LP)    | c1 :<br>J0148_PA200_rig<br>ht_v2                                                                                     | 3 Shutter Slitlet | 27.17142583333<br>334 Degrees<br>5.988976388888<br>89 Degrees | 200.00138485318<br>01    |                              |                                    | 3                          | 18                  | 26522.602                    |
|                   | 2                                                                                                                                                                                                                                                                                                                                                               | 1<br>(G140M/F070LP)    | c1 :<br>J0148_PA200_lef<br>t_v2                                                                                      | 3 Shutter Slitlet | 27.15064208333<br>334 Degrees<br>6.020551388888<br>88 Degrees | 199.99913236971<br>315   |                              |                                    | 3                          | 18                  | 26522.602                    |

Proposal 9886 - Observation 2 - Lyman-alpha tomography of a proto-cluster with a luminous quasar in the early universe

|                      |                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Special Requirements | Group Visits within 53.0 Days<br>Visits Same PA<br>MSA Planned Aperture PA 200.0000 to 200.0000 Degrees (V3 61.4254303 to 61.4254303) |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|